What Target-wise ABA Learning Does on Deterministic Causal Fixtures

Evidence from controlled probes H0–H7b

Working synthesis for discussion with Fabrizio

4 August 2026

Status and purpose. This is a supervisor-facing working document. It introduces the causal fixture, learner input, terminology, configurations, and experimental sequence before stating any cross-cutting findings. The controlled probes labelled H0–H7b are complete and analysed; the labels are internal shorthand only and each probe is described in the text by its scientific question and design. No cross-cutting conclusions are asserted yet. The six Findings headings and the Implications heading at the end are intentionally empty placeholders for subsequent drafting and review.

Abstract

The investigation asks what target-wise ABA Learning does when given only a finite binary table sampled from a known causal data-generating process. The principal process is a binary collider $A \to C \leftarrow B$ with independent stochastic roots and a deterministic AND child $C = A \wedge B$. The graph, mechanisms, population distribution, finite sample, learner-visible encoding, learned ABA framework, and evaluator-only reference are kept separate throughout.

The experimental sequence is recorded under labels H0–H7b for reference. It begins with a baseline comparison of the published Greedy and non-deterministic configurations across all targets on one frozen table (H0), then introduces one controlled question at a time: finite-sample support (H1); an irrelevant trailing covariate (H2); background-knowledge order with a leading distractor (H3); brave versus cautious learning (H4–H5); folding-token budget and grounded sample size (H6); and the `relto` versus `sechk` assumption-introduction options (H7). This document records what each probe was intended to reveal, its principal signal, and the procedural explanation supported by the generated deltas and Prolog traces.

Contents

1 Purpose, scope, and organising question

1.1 Research question

> For a finite binary table generated by independent stochastic roots and deterministic non-root mechanisms, and for each observed variable taken in turn as the learning target: what ABA framework does ABALearn emit under the configurations studied here, and how does that framework relate to the evaluator-only mechanism reference and to the information present in the population and in the sample?

The purpose is to describe how the learner transforms the supplied table into rules, assumptions, and contraries, and to classify each emitted theory against a target-dependent evaluator reference. On the principal AND collider $A \to C \leftarrow B$, only the deterministic child C has a desired learned rule over its observed parents:

```
c(A) :- a_val_1(A), b_val_1(A).
```

The stochastic roots A and B have evaluator reference `no_observed_parent_deterministic_rule`: the desired reading for those targets is that the learner emit no deterministic parent rule.

Relative to that target-dependent reference, the outcomes distinguished in this document are:

- for a deterministic child: a rule that coincides with the mechanism;
- for a deterministic child: a sample-adequate but non-minimal rule;
- for a deterministic child: a defeasible representation that differs from the mechanism string;
- for a root or child: a rule that holds on the finite sample because a population counterexample is missing from that sample;
- for a root: an assumption construction that satisfies the selected learning semantics while leaving the target underdetermined by its predictors;
- search termination under the configured procedure, including timeout or `completed_no_solution`, which for a root may be the desired diagnostic reading.

These classes separate procedural exit status, finite-sample adequacy, and mechanism alignment under the target-dependent reference. They are the vocabulary used when interpreting each probe below.

1.2 Scope

The evidence comes from three small binary fixtures, fixed seed-42 samples, and the learner configurations described below. Learning is target-wise: one frozen table is held fixed, and each observed variable is selected in turn as the learning target. Changing the target changes the positive and negative examples and the background-knowledge encoding of the remaining columns; the underlying rows remain the same sample.

Only the deterministic child has an evaluator-desired deterministic rule over its observed parents. Root-target runs are diagnostics of the same pipeline when that desired object is absent: they show how finite support, learning semantics, and search budget shape the emitted theory or search outcome for a stochastic root.

1.3 Object of study

The pipeline under study is target-wise ABA Learning over encoded tabular data. For each target it receives background knowledge and labelled examples only, and it emits an ABA framework of rules, assumptions, and contraries under a selected learning semantics and search configuration.

The evidence recorded here therefore concerns learned rule structure, search behaviour, and the relation of those outputs to the evaluator-only mechanism reference for the fixture under study.

2 Conceptual framework and notation

2.1 The seven objects that must not be conflated

The experimental reasoning separates the following objects:

1. a generating DAG G;
2. structural mechanisms for the variables in G;
3. the exact observational population distribution P^*;
4. a finite IID sample $\mathcal{D}_n \sim P^*$;
5. target-specific learner input derived from $\mathcal{D}_n$;
6. the learned ABA framework and its emitted delta; and
7. evaluator-side judgements about sample adequacy, mechanism alignment, minimality, and causal meaning.

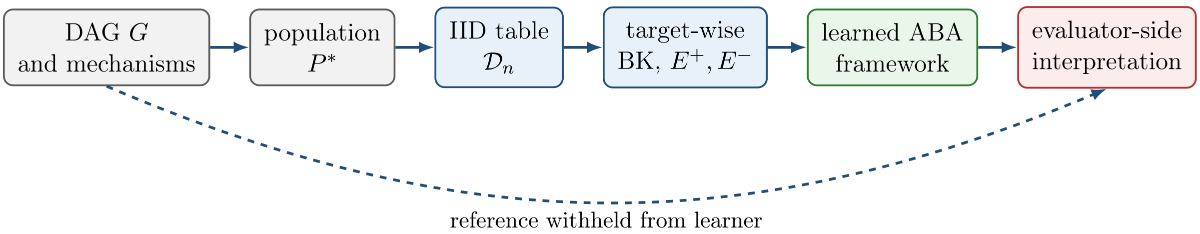

Figure 1: Experimental separation. Only the finite table and its target-wise encoding cross the learner boundary. The generating graph and mechanism reference are used only for evaluation.

These levels answer different questions. Faithfulness is a property of P^* relative to G, not of one finite table. A learned implication may hold on $\mathcal{D}_n$ but fail in P^*. A learned rule may mention a parent without establishing the direction of a causal edge.

2.2 ABA framework

An ABA framework is a tuple

$$\mathcal{F} = \langle \mathcal{L}, \mathcal{R}, \mathcal{A}, \bar{\cdot} \rangle,$$

where $\mathcal{L}$ is a language, $\mathcal{R}$ is a set of rules, $\mathcal{A} \subseteq \mathcal{L}$ is a set of assumptions, and $\bar{\cdot}$ maps each assumption to its contrary.

A rule has the form

$$h \leftarrow b_1, \ldots, b_m.$$

If $m = 0$, it is a fact. Assumptions are defeasible premises: an argument may use an assumption α, but deriving $\overline{\alpha}$ attacks it. A stable extension is, informally, a coherent set of accepted assumptions which attacks every assumption outside it. The learned framework can therefore represent an ordinary rule together with explicit exceptional conditions.

2.3 ABA Learning task

An ABA Learning task begins with background knowledge and two finite sets of examples:

$$E^+ = \{e_1^+, \ldots, e_p^+\}, \qquad E^- = \{e_1^-, \ldots, e_q^-\}.$$

The learner applies symbolic transformations to construct a framework $\mathcal{F}'$ under which the examples satisfy the selected consequence condition. The learned object is an ABA framework, not a fitted numerical parameter vector.

The transformations relevant here are:

- **Rote learning.** Add example-specific rules from the positive examples.
- **Folding.** Replace example-specific conditions with reusable BK predicates, producing intensional rules.
- **Subsumption.** Remove a specialised rule when an appropriate more general rule already exists.
- **Assumption introduction.** Make an over-general rule defeasible by adding an assumption; learn rules for its contrary to represent exceptions.

2.4 Exact-value target-wise encoding

For each row identifier i, every non-target variable is represented by exactly one value predicate. When c is the target, for example, the BK contains facts such as

```
a_val_1(9).
b_val_0(9).
```

and row 9 contributes either $c(9) \in E^+$ or $c(9) \in E^-$, according to its target value. For a binary target Y,

$$E_Y^+ = \{Y(i) : Y_i = 1\}, \qquad E_Y^- = \{Y(i) : Y_i = 0\}.$$

All non-target columns are available as BK predictors in their serialized order. This creates a symmetric target interface, but it does not make the variables causally symmetric: a deterministic child and a stochastic root have different evaluator-side recovery objects.

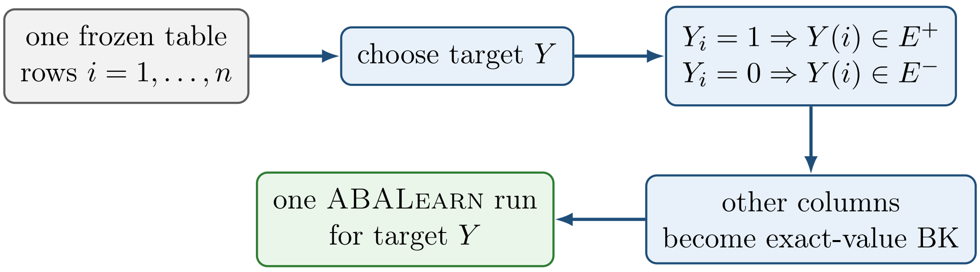

Figure 2: Target-wise encoding. All targets use the same rows; the selected target determines the positive and negative examples and the remaining columns determine the BK.

2.5 Brave and cautious learning

Let $\mathrm{Stab}(\mathcal{F}')$ be the stable extensions of the learned framework. Brave learning uses an existential witness: there must be some $\Delta \in \mathrm{Stab}(\mathcal{F}')$ which simultaneously accepts every positive example and rejects every negative example. In compact form,

$$\exists \Delta \in \mathrm{Stab}(\mathcal{F}') \quad \left[\forall e \in E^+,\ \mathcal{F}' \models_\Delta e \ \wedge\ \forall e \in E^-,\ \mathcal{F}' \not\models_\Delta e\right].$$

Cautious acceptance requires a claim to be accepted with respect to every stable extension. A cautious learning solution therefore requires all positives to be cautious consequences and requires each negative not to be a cautious consequence:

$$\forall e \in E^+,\ \mathcal{F}' \models_{\mathrm{caut}} e, \qquad \forall e \in E^-,\ \mathcal{F}' \not\models_{\mathrm{caut}} e.$$

This distinction matters whenever introduced assumptions permit more than one stable extension. Brave learning may use one joint witnessing extension. Cautious learning tests whether a positive survives every stable extension. Neither mode by itself establishes that a target is a deterministic function of the supplied predictors.

2.6 Greedy and non-deterministic folding

The configurations studied here use two different search strategies.

Greedy folding. The AAMAS configuration greedily folds a rote rule over the matching exact-value BK literals, with `mgr` selection and BK folding space. In the current complete exact-value representation, its solved Horn rules can become maximal co-occurrence descriptions. Folding is applied per rule; the learner does not subsequently synthesise a more general rule by deleting a literal across several complementary rules.

Non-deterministic folding. The nd configurations fold incrementally with `any` selection and the larger `all` folding space. A first admissible BK predicate can therefore become the first generalising fold. If that rule over-generalises, assumption introduction and contrary learning can preserve the initial fold while adding defeasible repair structure around it.

Greedy and nd configurations change a bundle of settings. A direct AAMAS-versus-ECAI comparison can reveal different behaviour, but it cannot attribute that difference to folding mode alone.

2.7 The `relto` and `sechk` assumption-introduction options

Both options are invoked when a folded candidate fails the relevant entailment check and the learner attempts to make the rule defeasible. They choose different procedural routes.

`asm_intro(relto).` The learner first searches for an assumption relative to the current folded body. In the inspected traces, a failed reuse can return KO and permit backtracking to a different folding literal. This is how later BK variables can enter contrary rules.

`asm_intro(sechk).` The learner first generates a new provisional assumption and tests the resulting framework through the `gen5`/`gen6` semantic-checking path. Depending on the learning semantics, the provisional assumption may close a repair early or may itself be rejected, forcing further nesting and search.

`relto` and `sechk` are implementation options, not causal criteria. A variable reached by one path is not thereby causal, and a shorter or longer learned theory is not automatically more correct.

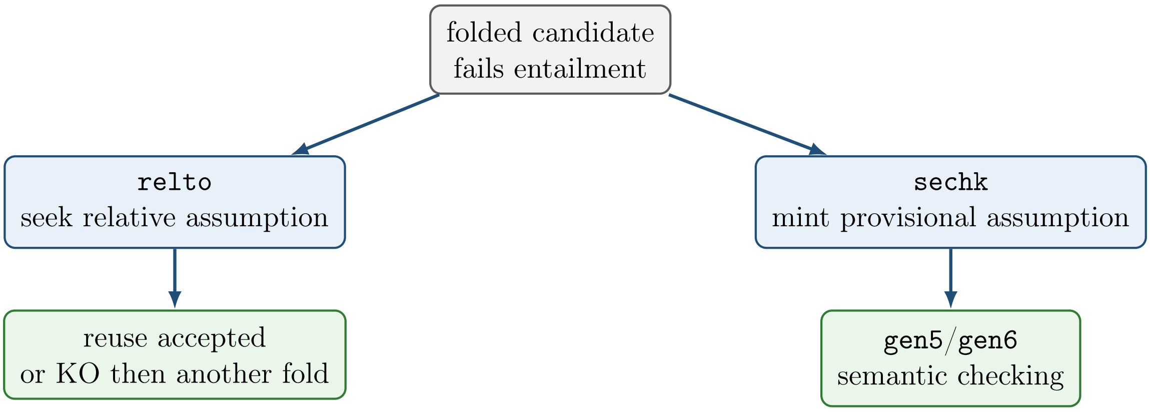

Figure 3: Repository assumption-introduction fork. H7 changes only this option within a fixed brave-nd or cautious-nd bundle. The subsequent path is also conditioned by the selected entailment semantics.

2.8 Evaluation vocabulary

Term	Meaning in this document
`solved`	The learner emitted an accepted framework under the selected configuration and semantics.
`completed_no_solution`	Search terminated without an accepted theory, relative to the configured strategy and resource ceiling.
`timeout`	The wall-time limit was reached; this is not evidence of unlearnability.
Sample-adequate	The framework satisfies the encoded finite examples under the chosen semantics.
Mechanism-aligned string	A learned rule is syntactically identical to the evaluator-only deterministic rule.
Distractor-retaining	A learned theory refers to a variable absent from the target mechanism.
Spurious finite-sample rule	A rule holds on the observed table but is false in the generating population.
Mechanism recovery	A justified conclusion that the learned theory represents the deterministic structural mechanism; stronger than `solved`.
Graph recovery	A justified conclusion about edges, orientations, an MEC, or a CPDAG; not produced automatically by any target-wise delta.

The generated `bk.sol_chk.asp` file is an integrity-check artefact for the serialized solution. Its satisfiability is not an independent graph-recovery metric and, under cautious learning, is not by itself proof that every positive is a cautious consequence.

3 Causal fixtures and evaluator ground truth

3.1 H0 baseline: binary AND collider

The principal fixture is

$$G_0 : \qquad A \longrightarrow C \longleftarrow B,$$

with learner identifiers a, b, c. The roots are mutually independent:

$$A \sim \mathrm{Bernoulli}(4/5), \qquad B \sim \mathrm{Bernoulli}(7/10), \qquad A \perp\!\!\!\perp B.$$

The child is deterministic:

$$C = A \wedge B.$$

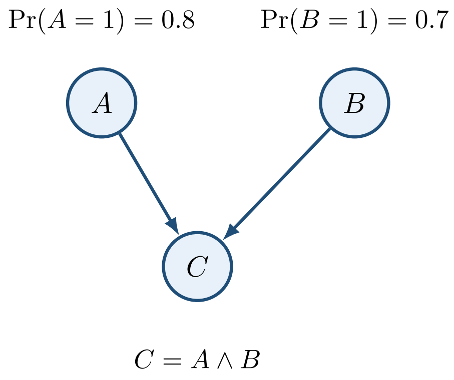

Figure 4: H0 generating DAG. Randomness enters through the independent roots; the non-root mechanism is deterministic.

The induced population distribution is:

A	B	C	$P^*(A, B, C)$
0	0	0	$3/50 = 0.06$
0	1	0	$7/50 = 0.14$
1	0	0	$6/25 = 0.24$
1	1	1	$14/25 = 0.56$

The remaining four binary assignments are structural zeros. Causal sufficiency is supplied by the fixture design: the root exogenous noise terms are mutually independent and there is no omitted common cause. The causal Markov condition follows from

$$P^*(a, b, c) = P^*(a)P^*(b)P^*(c \mid a, b).$$

Ordinary faithfulness was verified exactly by auditing every singleton-pair conditional-independence query against d-separation. The sole population independence is $A \perp\!\!\!\perp B$; conditioning on the collider makes $A \not\perp\!\!\!\perp B \mid C$, and each parent is dependent on C.

The skeleton is $A - C - B$ and the unshielded collider is $A \to C \leftarrow B$. Consequently, the ordinary conditional-independence MEC contains only this DAG and the CPDAG has both arrows directed into C. This population-level causal reference is never shown to ABALearn.

3.2 Evaluator-only mechanism reference

For H0, the canonical rule corresponding to the deterministic child mechanism is:

```
c(A) :- a_val_1(A), b_val_1(A).
```

Targets a and b are stochastic roots. Their evaluator reference is `no_observed_parent_deterministic_rule`; there is no desired rule that deterministically derives either root from the other observed variables.

Syntactic equality with the rule above is strong fixture-specific evidence, but not sufficient on its own for graph recovery. Conversely, a different ABA representation may be extensionally adequate without matching the canonical string. The records therefore inspect rule structure, assumptions, contraries, and the target's generating role separately.

3.3 Frozen H0 sample

The principal table is the seed-42 IID sample with $n = 30$:

Assignment	Count
$(1, 1, 1)$	17
$(1, 0, 0)$	7
$(0, 1, 0)$	4
$(0, 0, 0)$	2

All four population atoms appear. The target example counts are

$$a : 24^{+}/6^{-}, \qquad b : 21^{+}/9^{-}, \qquad c : 17^{+}/13^{-}.$$

For child c, all positive examples have $A = B = 1$. For root target a, the predictor cell $(B, C) = (0, 0)$ contains both target labels. In particular:

Row	A target	B predictor	C predictor	Class for target a
9	1	0	0	positive
26	0	0	0	negative

No deterministic rule over (B, C) can distinguish these two rows. Target b has the symmetric ambiguity. This pair later provides the clearest test of brave versus cautious assumption use.

3.4 H2 and H3 controlled extensions

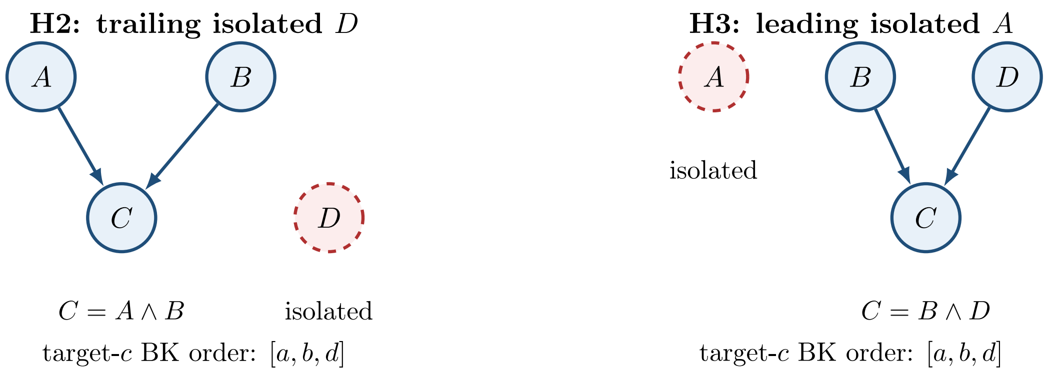

Figure 5: Controlled extensions. Dashed red nodes are independent isolated variables. H2 appends the distractor after the mechanism parents; H3 places it before them.

H2 preserves the H0 collider and AND mechanism and adds an isolated $D \sim \text{Bernoulli}(1/2)$. Its evaluator rule remains:

```
c(A) :- a_val_1(A), b_val_1(A).
```

H3 instead uses independent roots

$$A \sim \text{Bernoulli}(1/2), \qquad B \sim \text{Bernoulli}(4/5), \qquad D \sim \text{Bernoulli}(7/10),$$

with A isolated and $C = B \wedge D$. Its evaluator rule is:

```
c(A) :- b_val_1(A), d_val_1(A).
```

Both extensions have verified Markov factorisation and exact ordinary faithfulness certificates. The evaluator graph and rules remain hidden from the learner.

4 Experimental design and comparison logic

4.1 Learner configurations

Identity	Semantics	Folding	Selection	Space	Assumption introduction
`aamas2025`	brave	greedy	mgr	BK	`relto`
`ecai2024`	brave	nd	any	all	`relto`
`baseline_cautious`	cautious	nd	any	all	`relto`
`greedy_cautious`	cautious	greedy	mgr	BK	`relto`
`nd_brave_sechk`	brave	nd	any	all	`sechk`
`nd_cautious_sechk`	cautious	nd	any	all	`sechk`

`aamas2025` and `ecai2024` reproduce the relevant repository configurations associated with the corresponding published approaches. `baseline_cautious`, `greedy_cautious`, `nd_brave_sechk`, and `nd_cautious_sechk` are experimental identities used for controlled comparison. In particular, `greedy_cautious` is not a published *AAMAS cautious* method.

The H6 configurations match `baseline_cautious` except for `folding_steps`. H7a matches `ecai2024` except for `asm_intro`; H7b matches `baseline_cautious` except for `asm_intro`.

4.2 Strength of comparisons

Three levels of evidence are distinguished.

- **Exact paired option ablation.** Fixture, sample, encoding, and all relevant learner settings are fixed except one option. H4, H4b, H5, H6a, H7a, and H7b provide this form of comparison.
- **Controlled data or fixture intervention.** One designed property of the sample or fixture changes. H1, H2, H3, and H6b provide this form.
- **Bundle-level contrast.** Several learner settings change together. AAMAS versus ECAI, or Greedy-cautious versus baseline cautious nd, can reveal behavioural differences but cannot attribute them to one setting.

This hierarchy prevents a useful qualitative contrast from being misrepresented as a single-option causal result.

4.3 Probe sequence

Probe	Question	Controlled intervention	Principal observable
H0	What does the baseline learner do for the child and roots?	One H0 table; AAMAS and ECAI; all targets	Outcomes, deltas, and traces.

Probe	Question	Controlled intervention	Principal observable
H1	Does a rare support witness prevent a false root rule?	Nested $n = 25$ versus $n = 30$, same stream and AAMAS	Root rules and counterexample rows.
H2	Does Greedy omit an irrelevant trailing covariate?	Add isolated D after the H0 parents	Target-c rule bodies.
H3	What happens when an isolated covariate is first in BK?	H3 fixture with BK order $[a, b, d]$	First nd fold and final child theory.
H4	Does cautious nd block the brave root construction?	Brave to cautious under fixed nd/`relto`	Root outcomes and control child delta.
H4b	How does cautious checking change the H3 child repair?	Same H3 target c, semantics only	Exact contrary-rule difference.
H5	Is brave-to-cautious behaviour different inside Greedy?	Full 18-cell Greedy replication, semantics only	Outcomes, deltas, traces, and runtime.
H6a	Why are cautious-nd no-solution traces long?	$M \in \{1, 2, 5, 10\}$, fixed H0 $n = 30$	Token bands, trace length, runtime.
H6b	How does grounded sample size affect the same search?	$n \in \{30, 60, 90\}$, fixed $M = 2$	Search topology and per-band growth.
H7a	How does `sechk` differ from `relto` under brave nd?	Assumption option only; H0 and H3	Seven paired deltas and trace forks.
H7b	Does the same difference persist under cautious nd?	Assumption option only; H0 and H3	Child deltas, root cost, timeout limits.

5 Experimental sequence: H0–H7b

5.1 H0: establish baseline behaviour by target and strategy

Question and purpose. H0 asked what Greedy AAMAS and nd ECAI learn from the same complete-support $n = 30$ H0 table when each of a, b, c is used as the target. This established the first mechanism-aligned child result and exposed the different problem geometry of deterministic child and stochastic root targets.

Controlled comparison. The fixture, table, exact-value encoding, and target-specific examples were fixed. The two arms changed the full learner bundle: `aamas2025` used brave Greedy/`mgr`/BK folding, whereas `ecai2024` used brave nd/`any`/`all`. This is therefore a strategy-bundle contrast, not a folding-mode-only ablation.

Arm	Target	Outcome	Emphasised result
AAMAS	c	solved	Direct assumption-free AND conjunction.
AAMAS	a, b	no solution	Both-zero candidates conflict with observed opposite-label rows and repair dead-ends.
ECAI	c	solved	Unary $a = 1$ gate plus an assumption attacked by $b = 0$.
ECAI	a, b	solved	Brave assumption-choice construction on ambiguous predictor cells; no root mechanism recovered.

Emphasised observation. Greedy target c emits:

```
c(A) :- a_val_1(A), b_val_1(A).
```

This is string-identical to the evaluator mechanism. Nd target c, however, emits:

```
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).
assumption(alpha_1(A)).
contrary(alpha_1(A),c_alpha_1(A)) :- assumption(alpha_1(A)).
```

It represents c defeasibly: $a = 1$ supports c unless $b = 0$ attacks the required assumption. The theory is sample-adequate and cites both parents across its target and contrary rules, but it is not the canonical conjunction.

For target a, Greedy attempts the strict rules

```
a(A) :- b_val_1(A), c_val_1(A).
a(A) :- b_val_0(A), c_val_0(A).
```

The first is safe, while the second covers negative rows 26 and 30. Contrary learning reconstructs the same both-zero body, cannot introduce a further relative assumption, and returns no solution. Nd instead emits an assumption-rich theory containing:

```
a(A) :- alpha_2(A), b_val_0(A).
c_alpha_2(A) :- alpha_3(A), c_val_0(A).
c_alpha_3(A) :- alpha_2(A), b_val_0(A).
```

Trace-supported explanation. Rows 9 and 26 have identical predictor values $(B, C) = (0, 0)$ but opposite target-a labels. No deterministic predictor-to-target rule can distinguish them. Brave learning nevertheless needs only one stable extension that jointly satisfies the labelled examples. Its ground assumptions may take different statuses for different row identifiers, allowing the recursive α_2/α_3 construction to accept the positive and reject the negative. The resulting ungrounded theory does not define A as a function of (B, C).

Evidential boundary. H0 establishes that procedural `solved` can mean very different things. It does not show that Greedy is generally causally correct or that nd is generally misleading. The AAMAS/ECAI comparison changes a strategy bundle.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/learning_analysis.md`

5.2 H1: remove a rare support witness

Question and purpose. H1 asked whether AAMAS root no-solution depended on observing the rare $(0, 0, 0)$ population atom which contradicts the apparent both-zero root rule.

Controlled comparison. The fixture, population, seed sequence, AAMAS configuration, and encoding were fixed. H1 used the first 25 rows of the H0 seed-42 stream. H0 used the first 30. Thus $n = 25$ is an exact nested prefix, not an independent sample.

The H1 prefix contains $(1, 1, 1) : 14$, $(1, 0, 0) : 7$, and $(0, 1, 0) : 4$, but no $(0, 0, 0)$ row. The omitted atom has population probability 0.06.

Target	H0 $n = 30$	H1 $n = 25$	Evaluator reading
a	no solution	solved with two Horn rules	H0 desired; H1 spurious.
b	no solution	solved with mirror Horn rules	H0 desired; H1 spurious.
c	solved AND	solved AND	Control unchanged in kind.

Emphasised observation. Without the conflicting atom, AAMAS accepts:

```
a(A) :- b_val_1(A), c_val_1(A).
a(A) :- b_val_0(A), c_val_0(A).
b(A) :- a_val_1(A), c_val_1(A).
b(A) :- a_val_0(A), c_val_0(A).
```

The both-zero rules are false population implications. They are not recovery progress: the roots still have no deterministic mechanism over the remaining observed variables.

Trace-supported explanation. H0 and H1 traces follow the same Greedy route until candidate checking. H1 has no opposite-label witness in the relevant predictor cell, so the both-zero rule is accepted. H0 contains rows 26 and 30, so the rule covers negatives; assumption repair is attempted and then dead-ends. The outcome reversal is therefore attributable to observed support, not a change in the population mechanism or learner.

Evidential boundary. H1 is a support ablation, not evidence that $n = 30$ is generally sufficient or that larger n monotonically improves recovery. The two prefixes are not independent replicates.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h1_support_ablation.md`

5.3 H2: add an irrelevant trailing covariate

Question and purpose. H2 asked whether Greedy would omit a variable which is statistically and causally irrelevant to the deterministic child once both of its values occur among positive examples.

Controlled comparison. The H0 collider and AND mechanism were preserved. An independent isolated binary root D was appended after A, B in target-c BK order. The main run used $n = 30$, seed 42, under AAMAS. Among the positive c rows, both $D = 1$ and $D = 0$ occur nine times.

Emphasised observation. AAMAS solves target c, but emits two D-specific rules:

```
c(A) :- a_val_1(A), b_val_1(A), d_val_1(A).
c(A) :- a_val_1(A), b_val_1(A), d_val_0(A).
```

The pair covers the sample and jointly behaves like AND across the observed values of D. It is nevertheless non-minimal and fails to coincide with the D-free evaluator mechanism.

Trace-supported explanation. Greedy BK folding exhausts matching exact-value BK literals within each positive co-occurrence body. A $D = 1$ row therefore retains `d_val_1`; a $D = 0$ row retains `d_val_0`. Subsumption can remove a specialised rule only when a suitable more general rule already exists. There is no later cross-rule anti-unification or literal-deletion phase that combines the complementary rules into:

```
c(A) :- a_val_1(A), b_val_1(A).
```

Evidential boundary. The secondary $n = 2$ prefix was recorded only as supporting context. H2's scientific signal is the balanced $n = 30$ distractor split. One fixture does not prove retention for every possible encoding or Greedy problem.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h2_irrelevant_covariate.md`

5.4 H3: put an irrelevant covariate first in BK

Question and purpose. H3 asked how nd learning behaves when an isolated variable precedes the two mechanism parents in target-c BK order. It was designed to expose whether the first folding opportunity becomes structural scaffolding for the learned theory.

Controlled comparison. H3 uses $B \to C \leftarrow D$, $C = B \wedge D$, and an isolated root A. The target-c order is $[a, b, d]$. The ECAI brave-nd run is primary; an AAMAS run on the identical table supplies a bundle-level contrast.

Emphasised observation. The ECAI trace begins:

```
folding result: c(A) <- [a_val_1(A)]
```

and the final ordinary target rules retain the isolated variable as two outer gates:

```
c(A) :- alpha_1(A), a_val_1(A).
c(A) :- alpha_2(A), a_val_0(A).
```

Later contrary rules cite b and d, but the emitted delta does not coincide with:

```
c(A) :- b_val_1(A), d_val_1(A).
```

On the same sample, AAMAS emits:

```
c(A) :- a_val_1(A), b_val_1(A), d_val_1(A).
c(A) :- a_val_0(A), b_val_1(A), d_val_1(A).
```

Thus both learner bundles solve c while unnecessarily organising the theory around isolated A.

Trace-supported explanation. Nd starts with one folding token and `any` selection. In the serialized BK, an a-value literal is the first admissible fold, so the first generalisation is based on A rather than the mechanism parents. When the unary fold over-generalises, the learner introduces an assumption and learns contraries around that rule. Later predicates can repair the theory, but the initial a-gate remains in the ordinary target rule. Greedy reaches a different but related failure through maximal co-occurrence: it keeps every matching literal and splits the AND rule by the two values of A.

Evidential boundary. H3 establishes a representation-order effect on this fixture. It does not show that the first BK variable must appear in every possible nd solution, or that any particular statistical ordering would be causally correct.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h3_bk_leading_distractor.md`

5.5 H4: change brave nd to cautious nd

5.5.1 H4: roots on the H0 table

Question and purpose. H4 returned to the H0 row-9/row-26 ambiguity and asked whether the brave root solution survives when the learning consequence relation is cautious.

Controlled comparison. The H0 fixture, $n = 30$ table, target-specific examples, BK, nd folding, `any` selection, `all` space, `relto`, and folding ceiling were fixed. The only learning-relevant change was `learning_mode(brave)` to `learning_mode(cautious)`. The comparator is the locked ECAI brave collection; the cautious arm is named `baseline_cautious`, not ECAI.

Target	Brave nd/`relto`	Cautious nd/`relto`
a	solved	no solution
b	solved	no solution
c	solved	solved, byte-identical delta

Emphasised observation. For root a, the traces coincide through the early $\alpha_1, \alpha_2, \alpha_3$ construction. They diverge at the attempted closure:

```
c_alpha_3(A) :- alpha_2(A), b_val_0(A).
```

The brave trace records the α_2 reuse as OK and completes the theory. The cautious trace records the same reuse as KO, exhausts the configured token budget, and emits no delta. Target b mirrors this. Control target c still solves with a byte-identical delta.

Trace-supported explanation. The brave construction can exploit different grounded assumption choices for the conflicting row identifiers inside one witnessing stable extension. Cautious checking asks whether the required consequences survive every stable extension. The decisive residual reuse does not, so the root gadget is rejected. The unchanged child control shows that cautious semantics has not simply prohibited assumption-bearing theories; it blocks this particular ambiguous construction.

Evidential boundary. The cautious root result is relative to the tested configuration and ceiling $M = 10$. It does not prove that no cautious theory exists under all search strategies or budgets, and no-solution is not root-mechanism recovery.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h4_cautious_vs_brave.md`

5.5.2 H4b: child c on the H3 table

Question and purpose. H4b asked a narrower structural question: given H3's leading-a gate, how does cautious checking change the later b, d split when learning child c?

Controlled comparison. H3 fixture, sample, target c, BK order, nd settings, and `relto` were fixed. Only brave versus cautious learning changed.

Emphasised observation. Both arms solve. The leading-a gates and the α_1-nested versus α_2-flat contrary skeleton are shared. Exactly one closing rule changes:

```
% Brave:
c_alpha_3(A) :- alpha_1(A), a_val_1(A).

% Cautious:
c_alpha_3(A) :- d_val_1(A).
```

Trace-supported explanation. Cautious checking rejects reuse of α_1 on the $a = 1$ side, so search continues until `d_val_1` is accepted. The experiment therefore locates the semantic effect in the α_3 closure. It does not attribute the leading-a gates to cautious learning: those gates were already present in the H3 brave theory.

Evidential boundary. Neither solution recovers the direct B, D AND rule; both retain isolated A. H4b concerns the internal split and repair, not whether A disappears.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h4b_cautious_split_under_a.md`

5.6 H5: duplicate all Greedy runs under cautious learning

Question and purpose. H5 asked whether the brave-to-cautious change that mattered under nd would alter any completed Greedy result or trace, and whether Greedy-cautious no-solution searches incur the long replays seen under cautious nd.

Controlled comparison. Five completed deterministic Bucket 3 AAMAS collections were duplicated under experimental `greedy_cautious`: H0 $n = 30$, H1 $n = 25$, H2 $n = 30$, H2c $n = 2$, and H3 $n = 30$. This gave 18 exact target-wise pairs. `folding_mode(greedy)`, `mgr`, BK space, `relto`, data, BK, and examples were fixed; only the learning mode changed.

Emphasised observation. Across all 18 pairs:

- every outcome matches;

- every solved delta is byte-identical;
- the traces are isomorphic apart from entailment-logging surfaces;
- no new assumption or contrary structure appears; and
- cautious runtime is approximately 1.6–2.3× brave runtime, with mean ratio about 1.84, while remaining below one second.

The invariance includes the desirable H0/H1 child conjunctions, the H1 spurious root rules, the H2/H3 distractor-retaining child rules, and the H0 root no-solutions.

Trace-supported explanation. The inspected Greedy paths either accept strict Horn rules or fail before an assumption-bearing solution is produced. Consequently, changing how stable extensions are quantified does not change the emitted deltas on these cells. Greedy also terminates without the nd folding-token replay loop, explaining why its cautious root no-solutions remain short.

Evidential boundary. H5 establishes semantic inertness only across this completed 18-pair set. The large difference between Greedy-cautious and baseline cautious nd is a search-bundle difference, not an isolated consequence of `folding_mode`.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h5_greedy_cautious.md`

5.7 H6: explain cautious-nd search cost

5.7.1 H6a: folding-token ceiling

Question and purpose. H6a asked why H4's cautious root no-solution traces contained roughly 5,000 lines and took about one minute, and whether the configured `folding_steps(10)` ceiling was driving repeated work.

Controlled comparison. The H0 $n = 30$ sample and `baseline_cautious` bundle were fixed. `folding_steps` varied over $M \in \{1, 2, 5, 10\}$; the $M = 10$ collection was the existing H4 run and was not regenerated.

M	Target	Outcome	Runtime (s)	Trace lines	New assumptions
1	a	no solution	6.638	549	9
2	a	no solution	13.117	1047	18
5	a	no solution	32.775	2541	45
10	a	no solution	69.952	5031	90

Target b mirrors this. Target c solves at every ceiling within `tokens(1)`, with the same 134-line trace, approximately 1.3-second runtime, and byte-identical delta.

Emphasised observation. For the no-solution roots, trace length and runtime grow approximately linearly with M across the tested values. Each permitted token band adds about nine new assumptions and roughly 500 trace lines without producing a different accepted theory.

Trace-supported explanation. `folding_steps(M)` acts as a token-budget ceiling, not as a literal count of successful folds. The decisive root KO already occurs in the first band. When budget remains, the nd controller increments the token allowance and replays essentially the same failed assumption-introduction motif:

$$\text{lines}(M) \approx C_{\text{fixed}} + M\, C_{\text{failed band}}.$$

Through $M = 10$, the additional budget buys repeated failed search rather than a new explanation.

Evidential boundary. The experiment does not show that $M = 1$ is always sufficient. Other learnable tasks may require more tokens. It explains the tested root cost but does not yet provide a general stopping rule.

5.7.2 H6b: nested sample size

Question and purpose. H6b asked how cautious-nd trace size changes when the folding-token topology is held fixed but more grounded IID rows are supplied.

Controlled comparison. The H0 population, seed 42, `baseline_cautious` bundle, and $M = 2$ were fixed. Nested prefixes used $n \in \{30, 60, 90\}$.

n	a lines	a runtime (s)	NEW	KO	c lines
30	1047	13.117	18	32	134
60	1641	22.169	18	32	200
90	2241	32.291	18	32	266

All four H0 population atoms are already present at $n = 30$. Their multiplicities, ordered as $(0, 0, 0), (0, 1, 0), (1, 0, 0), (1, 1, 1)$, are:

$$\begin{aligned} n = 30 &: (2, 4, 7, 17), \\ n = 60 &: (4, 10, 12, 34), \\ n = 90 &: (5, 17, 18, 50). \end{aligned}$$

Emphasised observation. The root search topology stays fixed at 18 new assumptions and 32 KOs, while target-a trace length increases by 594 and 600 lines for each additional 30 rows. Target b mirrors this. Target c retains the same delta but its trace grows by 66 lines per 30 rows.

Trace-supported explanation. Additional rows create more grounded example identifiers, rote rules, entailment checks, subsumption work, and contrary batches inside each fixed search band. They do not introduce a new supported value pattern in this comparison. The approximately linear line growth therefore comes from per-row grounding and bookkeeping while the procedural topology remains unchanged.

Evidential boundary. H6b is a nested-n experiment, not a direct deduplication ablation. Multiplicities still carry empirical frequency information, and removing duplicate rows can alter row order and learner behaviour. The result is local to $n = 30, 60, 90$, not a universal complexity law.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h6_folding_and_n_ablation.md`

5.8 H7: change assumption introduction from `relto` to `sechk`

5.8.1 H7a: brave nd

Question and purpose. H7a asked whether the learned rules and trace path change when the assumption-introduction option is changed inside the brave nd bundle. It also asked whether any difference follows a consistent relationship to serialized BK order and causal role.

Controlled comparison. Within each pair, H0 or H3 fixture, $n = 30$ sample, target, examples, BK, brave semantics, nd folding, `any` selection, `all` space, folding ceiling, and other settings were fixed. The locked `ecai2024`/`relto` collection was compared with experimental `nd_brave_sechk`/`sechk`. There were seven target-wise pairs.

The target roles must be kept explicit:

Cell	Causal role	Evaluative use
H0/H3 c	deterministic child	desirable mechanism target
H0 a	parent of c, root as target	root diagnostic
H3 b	parent of c, root as target	role-peer of H0 a
H3 a	isolated root	leading-distractor diagnostic
H0 b, H3 d	other parent roots	supporting mirrors

Emphasised observation. Both arms solve all seven cells, but every paired `delta.aba` differs:

Cell	BK order	Outcome	`sechk` variables	Additional `relto` variables
H0 a	b, c	solved / solved	b	c
H0 b	a, c	solved / solved	a	c
H0 c	a, b	solved / solved	a	b
H3 a	b, c, d	solved / solved	b	c, d
H3 b	a, c, d	solved / solved	a	c, d
H3 c	a, b, d	solved / solved	a	b, d
H3 d	a, b, c	solved / solved	a	b, c

For H0 child c:

```
% relto
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).

% sechk
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- alpha_2(A), a_val_1(A).
c_alpha_2(A) :- alpha_1(A), a_val_1(A).
```

Both theories use the first a-predicate as the outer gate. `relto` later reaches mechanism parent b, whereas `sechk` closes an assumption chain around a. On H3 child c, `relto` reaches b, d, while `sechk` uses only the first, isolated variable a.

Trace-supported explanation. After the failed post-fold entailment check, `relto` first searches for a relative assumption. A KO can return control to folding and admit later BK literals. `sechk` instead generates a provisional assumption and enters `gen5`/`gen6`. Under brave semantics on these cells, the resulting assumption chain closes before the later folds reached by `relto`.

Every H7a `sechk` delta therefore cites only its first BK predictor block. This is not always variable a: for target a, the first predictor is b. Byte-identical H0/H3 `sechk` root deltas consequently reflect a shared learner template, not a shared causal role.

Evidential boundary. H7a does not establish that `sechk` is worse than `relto`, or that first-BK closure is universal. It records a bounded brave-nd interaction on seven exact input-matched pairs.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h7a_relto_vs_sechk.md`

5.8.2 H7b: cautious nd

Question and purpose. H7b asked whether the H7a `relto`/`sechk` difference transfers unchanged when both arms use cautious rather than brave learning.

Controlled comparison. H0 and H3 fixtures, frozen $n = 30$ tables, targets, examples, BK, nd folding, `any` selection, `all` space, and folding ceiling were fixed. Locked `baseline_cautious`/`relto` was compared with experimental `nd_cautious_sechk`/`sechk`. The comparator was not ECAI: both arms used cautious semantics.

Seven targets were run, but the evidence has unequal strength:

Question	Cell	Outcome pair	Evidential use
Q1	H0 c	solved / solved	Comparable deltas and traces.
Q2	H3 c	solved / solved	Comparable deltas and traces.
Q3	H0 b	no solution / no solution	Comparable failed-search cost.
Q4	H3 b	timeout / timeout	Non-comparable: both logs empty.
Q5	H3 a	timeout / timeout	Non-comparable: both logs empty.

Emphasised observation. Both cautious policies solve child c on both tables, but their paired deltas differ. For H0 c:

```
% baseline_cautious / relto
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).

% nd_cautious_sechk / sechk
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- alpha_2(A), a_val_1(A).
c_alpha_2(A) :- b_val_1(A).
```

`sechk` still inserts a new assumption layer around the first a-predicate, but cautious checking prevents the brave early closure and search eventually reaches b.

For H3 c, both policies retain the isolated-a gates and cite mechanism parents b, d somewhere in their contrary structures. The `sechk` theory is thicker:

Arm	Assumptions	Trace lines	Trace character
`relto`	3	257	Relative KO then a lean path to b, d.
`sechk`	5	899	Deeper nests and repeated NEW/`gen5`/KO cycles.

For H0 root b, neither cautious arm emits a delta. The `sechk` trace has approximately 13,818 lines, compared with 5,118 for `relto`, because it makes many more provisional-assumption and `gen5` attempts. H3 targets a and b time out in both arms with empty stdout files; those matched timeout labels reveal no trace-level similarity.

Trace-supported explanation. The same procedural fork as H7a occurs after failed post-fold entailment, but cautious semantics changes which provisional repairs survive. Relative reuse often receives a cautious KO and can expose another fold. The `sechk` path first adds a provisional assumption; when cautious checking rejects an early closure, further nesting and backtracking can eventually reach later b or d predicates. Thus H7b retains the H7a assumption-policy intervention while demonstrating that its observable effect is conditioned by brave versus cautious entailment.

Evidential boundary. The child theories remain non-canonical and, on H3, distractor-retaining. Mechanism-parent occurrence in a contrary is not mechanism recovery. The failed-root comparison concerns cost, and the empty-log H3 timeouts are non-comparable.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h7b_cautious_relto_vs_sechk.md`

6 Findings

6.1 Finding 1: Greedy solutions retain all represented BK predictors

6.2 Finding 2: nd solutions are organised around the first BK predictor

6.3 Finding 3: brave nd can solve predictor-ambiguous root targets

6.4 Finding 4: cautious-nd failed-search cost grows with the folding-token ceiling

6.5 Finding 5: cautious-nd trace size grows with grounded sample size

6.6 Finding 6: `relto` and `sechk` produce different assumption-introduction paths

7 Implications